

Towards AI-Mediated Support in Parenting Interventions: A Knowledge Graph Approach

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Our research examines how knowledge graph-enhanced retrieval-augmented generation can support AI-mediated collaboration in the development of parenting interventions. We investigate how different graph representations influence the system’s ability to support the curation and adaptation of evidence-based parenting intervention content, with implications for negotiation and alignment across stakeholders, positioning AI systems as mediators in the development of parenting interventions.

CCS Concepts: • **Applied computing** → **Document searching**; • **Information systems** → *Document structure*.

Additional Key Words and Phrases: Knowledge graphs, Evidence-based psycho-educational content, Artificial intelligence

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1 Introduction

Collaborative mental healthcare often involves ongoing negotiation between parents, young people, and clinicians, where differences in knowledge, expectations, and priorities can shape how care is understood and enacted. Parenting interventions play a critical role in supporting young people’s mental health; however, the effectiveness of such interventions depends not only on the quality of evidence, but also on how that evidence is translated into accessible, consistent, and engaging guidance for diverse families.

Web-based psycho-educational programs such as Partners in Parenting (PiP) [4] aim to bridge this gap by providing personalised, evidence-based resources for parents. Yet, curating and maintaining high-quality intervention content remains a complex and resource-intensive process. Experts must synthesise research findings, ensure consistency in tone and structure, adapt content for different contexts, and maintain alignment with evolving clinical guidelines. These tasks are not purely informational—they involve interpretation, trade-offs, and implicit negotiation between clinical evidence, user needs, and design considerations.

Recent advances in Generative AI offer new opportunities to support this process [3, 5]. However, most AI systems in healthcare focus on information retrieval or content generation without explicitly supporting the underlying need for structured reasoning, traceability, and alignment across stakeholders. This limits their ability to function as mediating agents that can assist experts in making informed, context-sensitive decisions during content development.

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To address this gap, this paper explores how Knowledge Graph-enhanced **Retrieval-Augmented Generation (RAG)** can support the curation and adaptation of parenting intervention content. By structuring domain knowledge into interconnected entities and relationships, **Knowledge Graphs (KGs)** enable more transparent and context-aware retrieval, allowing AI systems to better support both reference-driven and creative tasks [2]. We investigate two knowledge graph construction approaches—chunk-based and proposition-based—and analyse how they influence the system’s ability to assist experts in generating consistent, evidence-grounded, and contextually appropriate intervention content.

Through this work, we position AI not simply as a content generator, but as a potential mediator that supports collaborative sense-making and decision-making in the design of parenting interventions. We discuss implications for designing AI systems that facilitate alignment between clinical evidence, expert knowledge, and the needs of families in real-world healthcare contexts.

2 Modelling Methodology

To support AI-mediated content curation and adaptation, we develop a Knowledge Graph-based RAG system that structures intervention knowledge for more transparent and context-sensitive use. In this section, we illustrate the methodology for modelling psycho-educational information into two different knowledge graphs (Chunk-Based and Proposition-Based) and the retrieval format to support AI’s generation.

2.1 Parent Intervention Content Used for Modelling

The resulting knowledge graphs are constructed from 241 existing intervention documents originating from the PiP program [4], an award-winning web-based public health intervention designed to empower parents. Specifically, the chosen modules were: PiP Plus (<https://partnersinparenting.com.au/>), which supports adult carers in parenting skills to reduce adolescent risk of depression and anxiety, and PiP Kids-Autism (<https://pka.partnersinparenting.com.au/>), designed to help parents of autistic children lower their child’s risk of anxiety and depression. Its highly structured format enabling explicit representation of relationships between mental health concepts. This makes it a strong foundation for graph-based knowledge retrieval with high traceability. Additionally, preliminary analysis across existing PiP programs reveals recurring section patterns. This consistency highlights opportunities for an AI system built on top of these interventions to support the development of new interventions. The content is licensed under Creative Commons License (CC BY-SA 4.0), allowing for full utilisation without copyright concerns.

2.2 Knowledge Graph Construction Methods

A KG is structured as a graph $G = (V, E)$ consisting of a set of vertexes (or nodes) V and a set of edges E , representing entities and relationships derived from the knowledge sources, respectively. In this work, these sources are unstructured markdown files containing interconnected raw text, markdown syntax, and HTML tags. Therefore, extracting entities, relationships, and attributes from such unstructured data is a crucial step prior to KG construction. Although the two construction methods differ in their final graph structures, they share a common extraction process.

2.2.1 Extraction Process. The shared pipeline begins with a data loader that extracts text from PDF files (converted from raw markdown files), followed by a text splitter that segments the content into 500-character chunks with an overlap of approximately 100 characters. Each chunk is then embedded into a 756-dimensional vector space. The core stage involves extracting entities and relationships from the text. Guided by a predefined node and edge schema, we use

Gemini-2.5-Flash for this extraction process. To better control duplication within the KG, we then apply a rule-based entity deduplication step, where semantically similar entities above a defined threshold are merged along with their corresponding instances. Finally, the extracted knowledge is passed to the respective graph builders, and the resulting structure is written to the graph database (Figure 1).

2.2.2 Chunk-Based Graph. In this structure, the original intervention content is preserved through *Chunk* nodes that store the **exact text segments extracted from the source documents**. Source references, such as filenames, are represented as *Document* nodes, while each chunk maintains its sequential position through NEXT_CHUNK links and its parent *Document* through FROM_DOCUMENT relationships. Although splitting large documents into smaller chunks improves granularity, it can fragment related facts. To mitigate this, extracted *Entity* nodes are linked back to their originating chunks and connected through semantic relations, restoring cross-chunk context (Figure 2).

2.2.3 Proposition-Based Graph. Propositions, self-contained statements that each capture a single fact, have been shown to outperform retrieval based on larger units such as sentences or full passages. Their key advantage lies in providing a higher density of query-relevant information [1]. Accordingly, this graph structure stores **propositions rather than raw source text**, with only *Entity* nodes and their relationships retained from the source documents. Both nodes and edges include descriptive fields composed of extracted propositions, capturing either the nature of an entity or the semantic meaning of a relationship between two entities. To preserve traceability after removing *Chunk* and *Document* node types, each node and edge is provided with a field that aggregates the filenames of all contributing source documents (Figure 3).

2.3 Query Formulation

To evaluate AI-supported retrieval and generation workflows, we employ two complementary query types: **Referencing queries** and **Creative queries**. **Referencing queries** are designed to reflect tasks that directly engage with and reuse existing PiP content, such as locating prior definitions or checking how concepts were previously phrased to support continuity and consistency. In contrast, **Creative queries** are designed to probe concept-level reasoning and content adaptation, requiring the system to generate new content grounded in existing PiP principles, such as localising examples for different cultural contexts or re-framing guidance for new audiences. Each query represents a plausible task experts may encounter during evidence-based content development (Table 1). These tasks reflect points where experts negotiate between consistency, contextual adaptation, and stakeholder needs.

3 Discussion and Future Work

This work contributes to designing AI mediators that support alignment and shared decision-making in collaborative healthcare. We explore Knowledge Graph-enhanced Retrieval-Augmented Generation as a step towards AI systems that support mediation in parenting interventions. By structuring domain knowledge into explicit relationships, the approach enables more transparent, traceable, and context-sensitive content generation from clinical evidence.

Framing AI as a mediator highlights its role in supporting negotiation between evidence, expert interpretation, and family needs. In this context, AI can assist not only in content creation but also in fostering shared understanding across stakeholders. We have developed a prototype system and will evaluate it through participatory workshops with eight domain experts (ethics approved). The study will examine how experts engage with AI-generated content and its influence on decision-making. The findings will inform the design of AI-mediated systems that better support collaborative sense-making and negotiation in real-world healthcare settings.

While our current system supports experts during content development, we view this as an upstream form of mediation that shapes how guidance is ultimately interpreted and negotiated between parents, clinicians, and young people. Ultimately, we envision AI systems that do not simply generate content, but actively participate in mediating understanding and negotiation across stakeholders in collaborative healthcare.

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4 Appendices

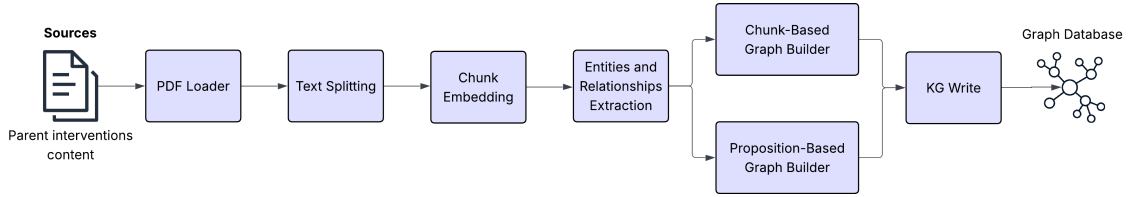


Fig. 1. Shared Extraction Pipeline

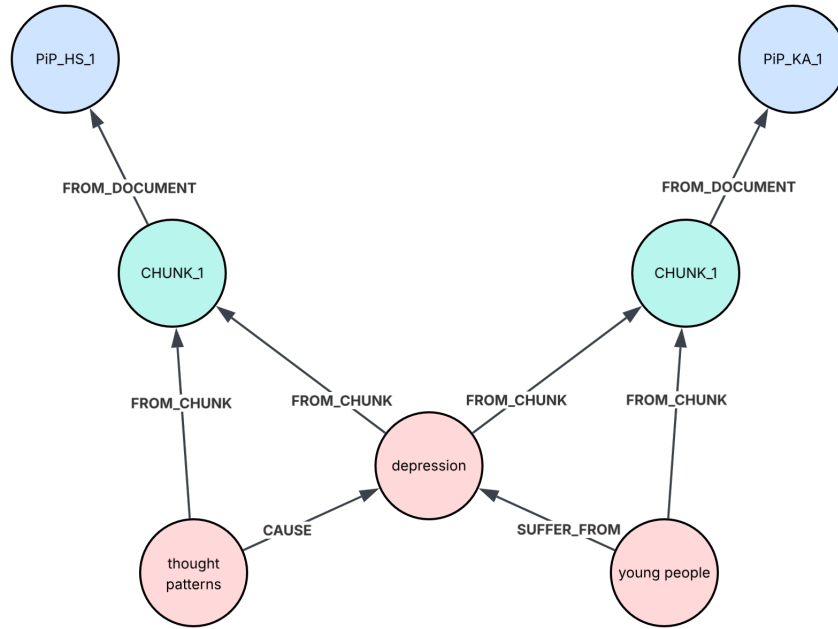


Fig. 2. Chunk-Based Graph

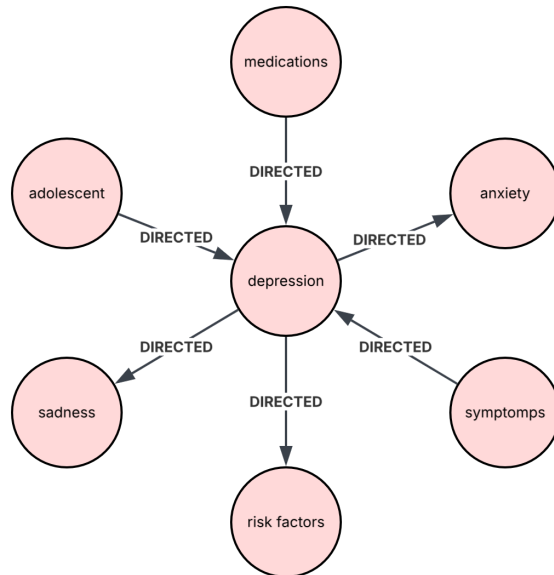


Fig. 3. Proposition-Based Graph

Query's Type	Query
Reference Task Query	What topics are covered in PiP that relate to anxiety in teenagers?
	How does PiP define emotional regulation in adolescents?
	Are there common principles across PiP modules about how parents should respond to children's emotional distress?
	Does PiP provide any frameworks or structured approaches for helping parents support teenagers experiencing anxiety?
	How does PiP typically describe and explain emotional distress in teenagers across different modules?
Creative Task Query	Generate a short scenario showing how a parent might respond supportively when a teenager expresses anxiety about school.
	Create a short example showing the difference between non-effective and effective parental responses to autistic children's communication difficulties and frustration.
	Based on concepts introduced in PiP, generate three practical tips for parents to help teenagers cope with exam-related stress.
	This is a conversation depicting a conflict between parent-teen: Parent: "I think you need to spend at least two more hours studying every night if you want to improve your grades." Teen: "Studying more hours isn't going to help if I'm already exhausted." Parent: "But your grades are slipping, so you need to put in more effort." Teen: "I am putting in effort. I just think you forcing more study time is just making me more stressed."
	Adapt this example so it is culturally appropriate for new migrant family from East Asia in Australia where open disagreement between parent-child is less common.
	The following PiP content refers to support mental health resources for parents. Replace these with equivalent services appropriate for a Canada context and explain differences for access: "1. If you would like to find a registered psychologist in your area, the Australian Psychological Society Find a Psychologist (https://psychology.org.au/find-a-psychologist) service can assist with this. You can also speak to your GP about getting a referral to a psychologist."

Table 1. Reference and Creative Task Queries